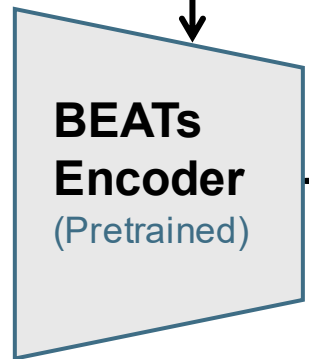
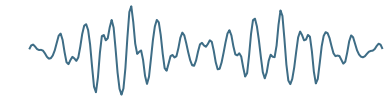


Respiratory Audio



H_i
 $T_i \times D$

Mean Pooling
(Linear $D \rightarrow d$)

h_i

Attention $\mathcal{G}$
(PAFA projection)

$\mathbb{R}^D$

Z_i

$\mathcal{L} =$

$\lambda_p \mathcal{L}_{PCSL}$

+

$\lambda_g \mathcal{L}_{GPAL}$

+

$\mathcal{L}_{cls}$

+

$\lambda_{HF} \mathcal{L}_{HF}$

Patient Contrastive

Patient Alignment

Native + C-W Classification

HF Supervision (Optional)

Task Heads

ICBHI Native Head
SPRSound Native Head

OR

Shared C-W Heads
C: Crackles
W: Wheezes

AND

Training Supervision

$\mathcal{L}_{cls}$

$\mathcal{L}_{HF}$

HF train* (Optional)

Task-Specific Outputs

ICBHI*:
Normal · Crackle · Wheeze · Both

SPRSound*:
Normal · Adventitious (Abnormal)

KAUH#:
Normal · Abnormal

HF test#:
Wheeze · Both

*: Training + Inference

#: Inference Only